

C21 ■ MULTISENSORY DESIGN | O (MAX : 4 PT)

Intent : Support neuro-inclusion through an informed planning process, multisensory design strategies and stakeholder education.

Summary : This WELL feature requires projects to facilitate a stakeholder charrette with a qualified neuro-inclusion professional, implement multisensory design strategies throughout the project and educate employees on neuro-inclusion.

Issue : Neurodiversity refers to the difference in neurocognitive profiles across an entire population.⁴ It is estimated that 15-20% of the world's population exhibits some form of neurodivergence.⁹ Human brains are diverse and vary; therefore, the way people interact with the environment can differ for each person.^{3,5} Neurocognition encompasses a wide range of profiles. This includes neurotypical, which describes someone fitting a majority neurological profile, as well as neurodivergent, which describes someone considered outside a majority or neurotypical neurological profile.^{3,9,20} People who are neurodivergent have unique sensory processing needs and may experience the environment at heightened or lower intensity levels.^{1,2,8} Companies actively hire people who are neurodiverse but few understand how to create supportive spaces.^{3,6,20} Traditional design often overlooks the specific needs of people who are neurodivergent as spaces are usually designed for neurotypical users.^{6,15} This leads to environments that are not supportive of people who are neurodivergent, which may hinder their ability to focus, cause anxiety and limit their participation.^{6,10,13} Although research is limited, there is a growing body of knowledge that is informing neuro-inclusive design.

Solutions : Due to the complex and nuanced nature of creating multisensory design strategies, it is important for qualified neuro-inclusion professionals and people who are neurodivergent to be engaged throughout the integrated planning and design process.^{2,9} Qualified professionals can help identify and implement evidence-based goals and solutions.² People who are neurodiverse can guide design decisions towards those that will be most important, impactful and supportive based on their lived experience.⁹ Creating spaces for neuro-inclusion also requires the implementation of multisensory design strategies to address individual differences in sensory processing. High-stimulation design strategies support hypo-sensitive individuals who may experience understimulation, while low-stimulation design strategies support hyper-sensitive individuals who may experience overstimulation.^{1,8,12} Individual control, access to nature, as well as acoustic and lighting elements are additional factors that support self-regulation and psychological safety for all individuals, regardless of how people process stimuli.^{1,7,8,17} In addition, ensuring that multisensory design strategies are sequenced can help provide natural transitions and a balance of sensory spaces throughout.^{1,11,12,13} Finally, stakeholder education can help communicate the benefits of neuro-inclusive design strategies and policies to all.^{3,4,10}

PART 1 PLAN FOR NEURO-INCLUSION (MAX : 1 PT)

For All Spaces:

Facilitate a stakeholder charrette that meets the following requirements:^{3,9}

- a. Is hosted early in the WELL planning process (may be part of the stakeholder charrette listed in Feature C02 Part 1).
- b. Involves a qualified neuro-inclusion professional.
- c. Includes efforts to involve representatives from the following stakeholders, as applicable:
 1. Owners.
 2. Organizational representatives who identify as neurodivergent.
 3. Interior design professionals (e.g., interior designer, facilities planner, lighting specialist, acoustics specialist).
 4. Human resources.
 5. Facilities managers.
 6. Architects.
 7. Engineers.
 8. Contractors.
- d. Accommodates participation from relevant stakeholders by addressing barriers (e.g., cultural norms or values, literacy levels, language, disabilities, work schedules, childcare) through timing, location, format and communication strategies.⁹
- e. Defines goals for how the project will support neuro-inclusion through multisensory design strategies.
- f. Identifies a list of credible sources (e.g., interviews, research studies, articles) that support how the identified neuro-inclusion goals may be addressed.
- g. Recommends a set of multisensory design strategies to support neuro-inclusion throughout the project.

Note :

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 CREATE SPACES FOR NEURO-INCLUSION (MAX : 2 PT)

For All Spaces:

1: Neuro-inclusive Design

Note : Projects may only achieve this part if Part 1 is also achieved.

The following requirements are met:

- a. At least five of the following multisensory design strategies are implemented:
 1. Adjustable environmental factors (e.g., screens, privacy panels, desks, chairs, lighting, temperature, acoustics) that allow people to control their sensory exposure are incorporated throughout.^{6,15,18}
 2. Natural forms, patterns, lines, shapes and objects (e.g., biomorphic, organic, fractal) are incorporated

throughout.^{5,17,15}

3. Environmental cues (e.g., visual, tactile, auditory, sensory maps) that indicate a shift in behavior or activity between spaces are incorporated throughout.^{2,16,19}
 4. Furniture that allows the body to stay in motion while using (e.g., rocking chair, balance boards).¹⁵
 5. Movable furniture that can be relocated without causing noise disruption (e.g., placed on soft floor, includes furniture pads) is incorporated throughout.⁶
 6. Enclosed areas that have interior windows providing clear lines of sight throughout the space are present.^{6,13,15}
 7. Walls that are curved or have rounded corners are present.^{1,13}
 8. Single-loaded corridors (i.e., doors to rooms are only located on one side only) are present.^{1,13}
 9. Personal storage for each regular occupant is available.¹⁵
 10. At least two entrances to the building are available, one designed with low-stimulation design strategies and one designed with high-stimulation design strategies.
- b. At least four of the following low-stimulation design strategies are implemented in at least two spaces:
1. Environmental factors that support visual, auditory and physical privacy.^{5,11,18}
 2. Areas of refuge.^{1,11,19}
 3. Low-intensity colors and patterns.^{1,5,13}
 4. Low-threshold textural experiences (e.g., materials that are smooth or predictable in typography, carpets with low pile).^{1,11,18}
 5. Restorative elements (e.g., biophilic patterns, calming sounds, horizontal lines).^{13,18,19}
 6. Use of intimate scale (e.g., low ceilings, tight spatial layout).^{6,13,19}
 7. Symmetrical organization of spatial layout and design elements.^{1,13}
- c. At least four of the following high-stimulation design strategies are implemented in at least two spaces:
1. Environmental factors that support visual, auditory and physical stimulation.^{13,15,19}
 2. Active and energizing design elements (e.g., natural sounds, low music, accent lighting).^{6,13,19}
 3. High-intensity colors and patterns.^{6,13,19}
 4. High-threshold textural experiences (e.g., materials that are rough or unpredictable in typography, carpets with high pile).^{6,13,19}
 5. Use of open scale (e.g., high ceilings, open spatial layout).^{6,13,19}
 6. Asymmetrical organization of spatial layout and design elements.^{1,13}
- d. Low-stimulation and high-stimulation spaces are distributed according to the following:
1. Interspersed throughout the project boundary.^{1,13}
 2. Incorporated in both work spaces and non-work spaces.

2: Flexible Work Space Policy

The following requirement is met:

- a. A flexible work space policy is in place that allows employees to work in different spaces within the project boundary and throughout the day based on their individual sensory needs.^{6,10,19}

Note :

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 3 EDUCATE FOR NEURO-INCLUSION (MAX : 1 PT)

For All Spaces:

1: Neuro-inclusive Employee Communication

The project meets the following requirements:

- a. Identifies and communicates the neuro-inclusive design strategies and related policies to employees through at least two of the following:
 1. Virtual tours, available on-demand.

2. In-person tours, available upon request.
3. Employee handbook.
4. Signage throughout the project boundary.

2: Neurodiversity Education

The project offers educational programming that meets the following requirements:

- a. Is provided by a qualified neuro-inclusion professional.
- b. Is offered annually to all employees.
- c. Is offered live, either in-person or virtually.
- d. Covers the following topics:
 1. Overview of the conditions of people who are neurodivergent.
 2. Preferred terms, language and definitions used to describe people who are neurodivergent.
 3. Evidence-based strategies to support and accommodate neuro-inclusion.
 4. Benefits of neuro-inclusive workplaces.
 5. Evidence-based strategies for preventing, identifying and navigating observed or experienced discrimination toward people who are neurodiverse.

Note :

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

REFERENCES

1. Cassidy, M. K. (2018). Neurodiversity in the Workplace: Architecture for Autism [Master's thesis, University of Cincinnati]. OhioLINK Electronic Theses and Dissertations Center. [Reference](#)
2. Cieslik, Emma. "Accessibility and Exhibit Safety: The Importance of Sensory Maps." Collections (2024): 15501906241232310.
3. Doyle, Nancy. "Neurodiversity at work: a biopsychosocial model and the impact on working adults." British medical bulletin 135, no. 1 (2020): 108-125.
4. Hedley, Darren, Sarah Cassidy, Chris Templin, Susan Hayward, Erin Bulluss, Jac den Houting, Kairi Kolves et al. "Recommendations from the 2021 Australasian Society for Autism Research "Health, Wellbeing and Suicide Prevention in Autism" Conference and Roundtable." (2022).
5. Hewitt, Jean. "Neurodiversity & the Built Environment - PAS 6463:2022." BSI, 2022. Downloaded from: <https://www.bsigroup.com/en-GB/insights-and-media/insights/brochures/pas-6463-design-for-the-mind-neurodiversity-and-the-built-environment/>. Accessed July 7, 2024.
6. HOK, Designing a neurodiverse workplace. (2019). Downloaded from <https://www.hok.com/ideas/publications/hok-designing-a-neurodiverse-workplace/>. Accessed October 26, 2021.
7. International Association of Accessibility Professionals. "Certified Professional in Accessible Built Environments (CPABE)." International Association of Accessibility Professionals. Downloaded from: <https://www.accessibilityassociation.org/s/certified-professional-environments>. Accessed July 7, 2024.
8. Kwon, J., Linihan, S., Iedema, A., Schmidt, A., Luo, C., & Marrufo, K. (2023). How interior design responds to neurodiversity: implementing wearable technologies in neurodesign processes. *Frontiers in Built Environment*, 9, 1211519.
9. Le Cunff, A. L., Logan, P. E., Ford, R., Martis, B. L., Mousset, I., Sekibo, J., ... & Giampietro, V. (2023). Co-design for participatory neurodiversity research: collaborating with a community advisory board to design a research study. *Journal of Participatory Research Methods*, 4(1).
10. LeFevre-Levy, Rose, Arturia Melson-Silimon, Rebecca Harmata, Anna L. Hulett, and Nathan T. Carter. "Neurodiversity in the workplace: Considering neuroatypicality as a form of diversity." *Industrial and Organizational Psychology* 16, no. 1 (2023): 1-19.
11. Mostafa, Magda. "Architecture for autism: Built environment performance in accordance to the autism ASPECTSS design index." In *Autism 360°*, pp. 479-500. Academic Press, 2020.
12. Mostafa, Magda. "The autism friendly university design guide." Dublin City University, Dublin (2021).

13. Mostafa, Magda, Marlene Sotelo, Toby Honsberger, Christine Honsberger, Erin Brooker Lozott, and Nate Shanok. "The impact of ASPECTSS-based design intervention in autism school design: a case study." *Archnet-IJAR: International Journal of Architectural Research* 18, no. 2 (2024): 318-339.
14. Paron, A. J. "Design for all needs design empathy." *Journal of Interior Design* 45, no. 4 (2020): 3-9.
15. Paron-Wildes, A. J. *Interior Design for Autism from Adulthood to Geriatrics*. John Wiley & Sons, 2013.
16. Patel, Tina, Juliann Dorff, and Allison Baker. "Development of special needs classroom prototypes to respond to the sensory needs of students with exceptionalities." *Archnet-IJAR: International Journal of Architectural Research* 16, no. 2 (2022): 339-358.
17. Rita Trombin. *Working with fractals: A resource for practitioners of biophilic design*. New York: Terrapin Bright Green, 31 December 2020.
18. Tarkett, *Sensory Processing Neurodiversity and Workplace Design* (2022). Downloaded from: [Reference](#). Accessed February 1, 2023.
19. Weber, Clara, Beate Krieger, Eunji Häne, Joanna Yarker, and Almuth McDowall. "Physical workplace adjustments to support neurodivergent workers: A systematic review." *Applied Psychology* 73, no. 3 (2024): 910-962.
20. HOK, *Designing neuroinclusive laboratory environments*. (2024). Downloaded from <https://www.hok.com/ideas/publications/designing-neuroinclusive-laboratory-environments/>